

Zachary L. Strasberg

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EDUCATION

The University of New Mexico, Albuquerque, NM

Ph.D. Student in Earth & Planetary Sciences (Expected 2027)

M.S. in Earth & Planetary Sciences (2024)

James Madison University, Harrisonburg, VA

B.S. with Honors in Geology, Minors in Statistics & Mathematics, Magna Cum Laude (2021)

TECHNICAL SKILLS

Machine Learning Modeling and Data Analysis: Statistical and machine learning techniques for climate / environmental research, including regression (XGBoost, Random Forest), time series forecasting (ARIMA, LSTMs, ReservoirPy), image analysis (RCNN), PCA, and cluster analysis

Computational and Geospatial Tools: Python (Pandas / GeoPandas / Xarray / SciKit-Learn / PyTorch), R (Tidyverse), ESRI Products (ArcPro / ArcMap / ArcGIS Online), HEC-HMS, JavaScript (D3 / DC / Bootstrap / NodeJS / Leaflet), HTML / CSS, SPSS, Stata, Microsoft Excel.

SELECTED PROJECTS

Threat, Race, Ability, Position, and Score (TRAPS) Backgammon Project:

Scraped and batch-analyzed high-level backgammon matches from itikawa.com using Python and eXtreme Gammon; parsed JSON outputs to evaluate a custom doubling formula developed by a published author. Built scorecard for visualization of TRAPS formula.

Mars Crater Identification: Co-developed a convolutional neural network to identify rampart craters on Mars using image data from remote sensing sources.

Mars Crater Web-App: Created a JavaScript web-app that visualizes a Mars crater database.

RELEVANT EXPERIENCE

Research Assistant, University of New Mexico (2021-Present)

- Develops drought prediction and attribution models using statistical and machine learning techniques applied to reanalysis datasets; collaborates with Sandia National Laboratories.

Faculty Assistant, University of Maryland / NOAA NCEI NESDIS (2024-Present)

- Updates MySQL/MariaDB relational databases with Python scripts for NOAA's Scientific Computing System, ensuring accurate oceanographic metadata.

Lapenta Intern, NOAA Ocean Acidification Program (2023)

- Developed geospatial analysis tools and conducted observing system simulation experiments (OSSEs) to assess ocean geochemistry measurements and interpolation models.

Data Analysis Intern, ADI Compliance Consulting (2018-2023)

- Performed quantitative analysis of large financial datasets to evaluate regulatory compliance and inform data-driven decision-making in financial risk assessment.

SELECTED PUBLICATIONS AND PRESENTATIONS

Co-Author: Polyak V, Asmerom Y, Curry B, **Strasberg Z** *et al.* (2023) *Geology*: 51 1048-1052.

Co-Author: Fahad A, Burls N, **Strasberg Z** (2020). *Climate Dynamics*: 55 703-718.

Poster: **Strasberg Z** *et al.* (2024). Developing Machine-Learning and Statistical Models for Western US Seasonal Precipitation. *American Geophysical Union Fall Meeting #A51L-1838*.

REFERENCES

Available upon request.